The link between erectile and cardiovascular health: the canary in the coal mine.

Lifestyle and nutrition have been increasingly recognized as central factors influencing vascular nitric oxide (NO) production and erectile function. This review underscores the importance of NO as the principal mediator influencing cardiovascular health and erectile function. Erectile dysfunction (ED) is associated with smoking, excessive alcohol intake, physical inactivity, abdominal obesity, diabetes, hypertension, and decreased antioxidant defenses, all of which reduce NO production. Better lifestyle choices; physical exercise; improved nutrition and weight control; adequate intake of or supplementation with omega-3 fatty acids, antioxidants, calcium, and folic acid; and replacement of any testosterone deficiency will all improve vascular and erectile function and the response to phosphodiesterase-5 inhibitors, which also increase vascular NO production. More frequent penile-specific exercise improves local endothelial NO production. Excessive intake of vitamin E, calcium, l-arginine, or l-citrulline may impart significant cardiovascular risks. Interventions discussed also lower blood pressure or prevent hypertension. Certain angiotensin II receptor blockers improve erectile function and reduce oxidative stress. In men aged <60 years and in men with diabetes or hypertension, erectile dysfunction can be a critical warning sign for existing or impending cardiovascular disease and risk for death. The antiarrhythmic effect of omega-3 fatty acids may be particularly crucial for these men at greatest risk for sudden death. In conclusion, by better understanding the complex factors influencing erectile and overall vascular health, physicians can help their patients prevent vascular disease and improve erectile function, which provides more immediate motivation for men to improve their lifestyle habits and cardiovascular health.